

**Volve
15/9-F-9
Mærsk Inspirer
USIT- CBL
13 3/8 inch
June 2009
Cement and formation evaluation**

Title: Volve 15/9-F-9 Mærsk Inspirer USIT- CBL 13 3/8 inch June 2009 Cement and formation evaluation		
Document no. :	Contract no.:	Project:

Classification: Internal	Distribution: Corporate Statoil
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Author(s)/Source(s): Aarti W. Paulus	
Subjects: Cement and formation evaluation from the received USIT and CBL logs taken in June 2009	
Remarks:	
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Techn. responsible (Organisation unit): TPD TEX PTEC PT PTDS	Technical responsible (Name): <i>AP</i> Aarti W. Paulus	Date/Signature: <i>Eirik Berg 17-15</i>
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Approved by (Organisation unit): TPD TEX PTEC PT PTDS	Approved by (Name): Linda Storetvedt Lien	Date/Signature: <i>16/11-15 Linda S. Lien</i>

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1 General Information

1.1 Well Information

Country	Norway	
Field Name	Volve	
Well Name	15/9-F-9	
Rig Name	Mærsk Inspirer	
Mud Details		
Mud type & weight in well at the time of logging (SG)	Seawater with 1.03 SG	<i>Brine</i>
Clean out run before logging (after cementing)	Yes	<i>6 hours before logging (26.06.2009)</i>
Casing Size (inch)	13.38	
From (mMD)	165.70	<i>Seabed</i>
To (mMD)	909.80	
Casing Weight (lb/ft.)	72	
Nominal ID (in)	12.35	
Inner radius (in)	6.17	
Nominal Thickness (in)	0.51	
External radius (in)	6.69	
Previous Casing Shoe Size (in.) and Depth (mMD)	30 220.8	
Max Deviation in Logged Interval (deg.)	11.6	

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1.1.1 Casing Summary from DBR

Casing Report

Wellbore: NO 15/9-F-9



Report Date: 27.11.2007 to 04.07.2009

Category/String type			Diameters		RKB hanger m MD	Air gap m MD	From depth m MD	To depth m MD	Date	Description	Tally
Conductor			30"			54,9	139,7	220,8	29.11.2007		
Item type	No. of joints	Diameter inch	Grade	Coupling	Weight lbs/ft	From depth m MD	To depth m MD	To depth m TVD	Remarks		
Casing	1	30"	X-65	ABB RL-4HC	456,600	139,7	152,8	152,8	Float collar: 0 mMD, Stage tool: 0 mMD		
Casing	6	30"	X-52	ABB ST-2 FB	309,700	152,8	220,8	220,8	Float collar: 0 mMD, Stage tool: 0 mMD		

Category/String type			Diameters		RKB hanger m MD	Air gap m MD	From depth m MD	To depth m MD	Date	Description	Tally
Casing, Surface			13 3/8"		139,7	54,9	165,7	909,8	15.12.2007		
Item type	No. of joints	Diameter inch	Grade	Coupling	Weight lbs/ft	From depth m MD	To depth m MD	To depth m TVD	Remarks		
Casing	3	13 3/8"	P-110	Vam TOP	72,000	165,7	201,7	201,7	Float collar: 883 mMD, Stage tool: 0 mMD		
Casing	59	13 3/8"	13 Cr-80	Vam TOP	72,000	201,7	909,8	905,1	Float collar: 883 mMD, Stage tool: 0 mMD		

Category/String type			Diameters		RKB hanger m MD	Air gap m MD	From depth m MD	To depth m MD	Date	Description	Tally
Liner, Production			9 5/8"			54,9	871,0	1 064,0	04.09.2008		
Item type	No. of joints	Diameter inch	Grade	Coupling	Weight lbs/ft	From depth m MD	To depth m MD	To depth m TVD	Remarks		
Casing	3	9 5/8"	13 Cr-80	Vam TOP	47,000	871,0	907,0				
Casing	14	9 5/8"	13 Cr-80	Vam TOP	47,000	907,0	1 064,0		Bakerweld 250 HD		

Category/String type			Diameters		RKB hanger m MD	Air gap m MD	From depth m MD	To depth m MD	Date	Description	Tally
Completion string (tubing)			7 5/8"			54,9	21,0	834,4	05.09.2008		
Item type	No. of joints	Diameter inch	Grade	Coupling	Weight lbs/ft	From depth m MD	To depth m MD	To depth m TVD	Remarks		
Tubing	66	7 5/8"	13 Cr-80	Vam TOP SC80	29,700	21,0	834,4				

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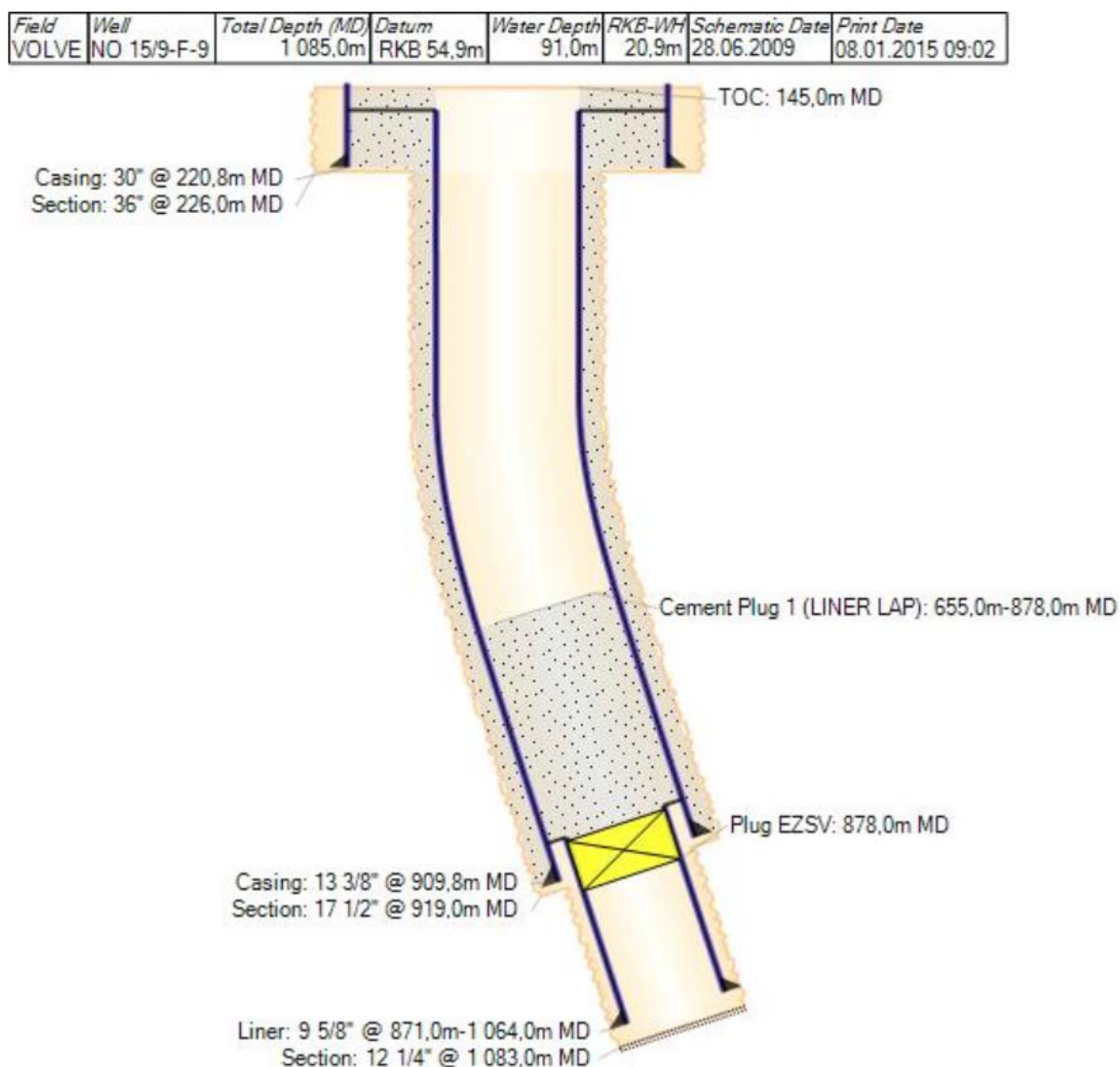


Figure 1: Wellsketch F-9 (Source: DBR)

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1.2 Logging Objectives

Cement evaluation.

Electrical Wireline Logging



Wellbore: NO 15/9-F-9

Report Date: 28.11.2007 to 04.07.2009

Run no	Log no	Report date	Logging company	Interval upper MD[m]	Interval lower MD[m]	Tools	Temp tool MD[m]	BHCT temp [degC]	BHST temp [degC]	Time since circ stopped	Remarks:
1	1	27.06.2009	Schlumberger	0,0	845,0	CBL - USIT					Logged the interval to check for good cement.
2	2	27.06.2009	Schlumberger	0,0	845,0	USIT - CBL					Repeat run for run no.1.

Figure 2: Electrical Wireline Logging report from DBR stating the objective.

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1.3 Cement Job Information

Cement Job Date and time when cement displacement was completed	16.12.2007	<i>Logged:26.09.2009</i>
Cementing Details		
Cement Type	Norcem G	
Cement Density	1.92 sg (tail) / 1.52 sg (lead)	
Cement Volume Pumped	15 m ³ (tail) / 104 m3 (lead)	
Expected Cement Impedance at time of logging	4 – 6 Mrayls (lead / tail)	
Maximum Theoretical Top of Cement	709 mMD (tail) – total job: seabed	<i>CHLG calculation based on actual cement volumes pumped and a gauge hole</i>
Mud weight in the well at the time of cementing	1,4 SG	<i>WBM (Bentonite)</i>
Cementing Operations		
Losses during cementing	No	
Any other issues during cementing operations	Return to seabed during cementing observed with ROV	

1.3.1 Cementing Report Summary from DBR

Casing string 13 3/8"		Cementing method Conventional		Company Halliburton			
Theoretical TOC MD 145,0 m	Evaluated TOC MD 145,0 m	Evaluated by	Liner rotation planned No	Liner rotation achieved		Bumped and successfully tested plug	
Remarks							
Fluids pumped	Type	Density g/cm3	Volume m3	Pump rate l/min	Pump pressure bar	Loss prior to cement job m3	Loss during cement job m3
Lead	Cement Slurry HTL50	1,52	40,0	1 200	26,00		
Lead #2	Cement Slurry Non Standard	1,52	62,0	1 200	24,00		
Tail	Cement Slurry MPT14	1,92	15,0	900	27,00		

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1.3.2 Cementing details

16.12.2007 19:15	16.12.2007 21:00	1,75	909,8	DP	U	CASING - CEMENT CASING/LINER Mixed and pumped 40 m3 1.52 sg lead 1 slurry at 1200 lpm / 28 bar followed by 61,8 m3 1.52 sg lead 2 slurry at 1200 lpm / 28 bar and 15 m3 1.92 sg tail slurry at 900 lpm / 28 bar. Observed with ROV returns up F-9 only.
16.12.2007 21:00	16.12.2007 21:45	0,75	909,8	DP	U	CASING - CEMENT CASING/LINER Released top dart. Displaced and sheared sub sea wiper plug by pumping 3 m3 SW with cement unit at 1200 lpm / 10 bar. Top plug sheared after 900 ltrs / 150 bar. Zeroed counters for mud pumps and lined up to displace cement with SW using mud pumps. Displaced cement with SW at 3000 lpm. Slowed down to 1200 lpm after 2250 stks. FCP was 40 bar. Bumped plug after 2505 strokes with 128 bar. Pump eff. 96.7%. Held pressure at 128 bar / 10 minutes to test casing. Observed with ROV returns up F-9 only.
16.12.2007 21:45	16.12.2007 22:30	0,75	909,8	DP	U	CASING - OTHER CEMENTING TIME, OK Bled of pressure and checked for backflow to cement unit. Volume bled back was 850 ltr. Disconnected cement hose after pressure had been bled off.

1.4 Logging Operations

Logging Passes	Date and Time	From (mMD RKB)	To (mMD RKB)	Log Resolution
Main Pass	27.06.2009	150	845	UWKM: 5 deg. 1.5 in.
Repeat Pass	27.06.2009	150	845	UWKM: 5 deg. 1.5 in.

Tool String	USIT-CBL/VDL-GR-CCL	USRS-D 809
Tractor Used	No	
Surface Pressure Applied During Logging (if yes how much in Bar)	No	

Other Comments:

- *Logged till 845 mMD*

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2 Log Quality Control

2.1 Wellbore Fluid Properties Parameters

Fluid Velocity Range in Main Pass Interval (us/ft.)	196.61- 207.66	Upper part is in riser
Theoretical ZMUD Range in Main Pass Interval (Mrayls)	1.60 – 1.5	Upper part is in riser – casing starts @ 140 mMD RKB
Mud Impedance Range used in final data on which the interpretation is based (Mrayls)	1.60	See figure 3 for more details - ok

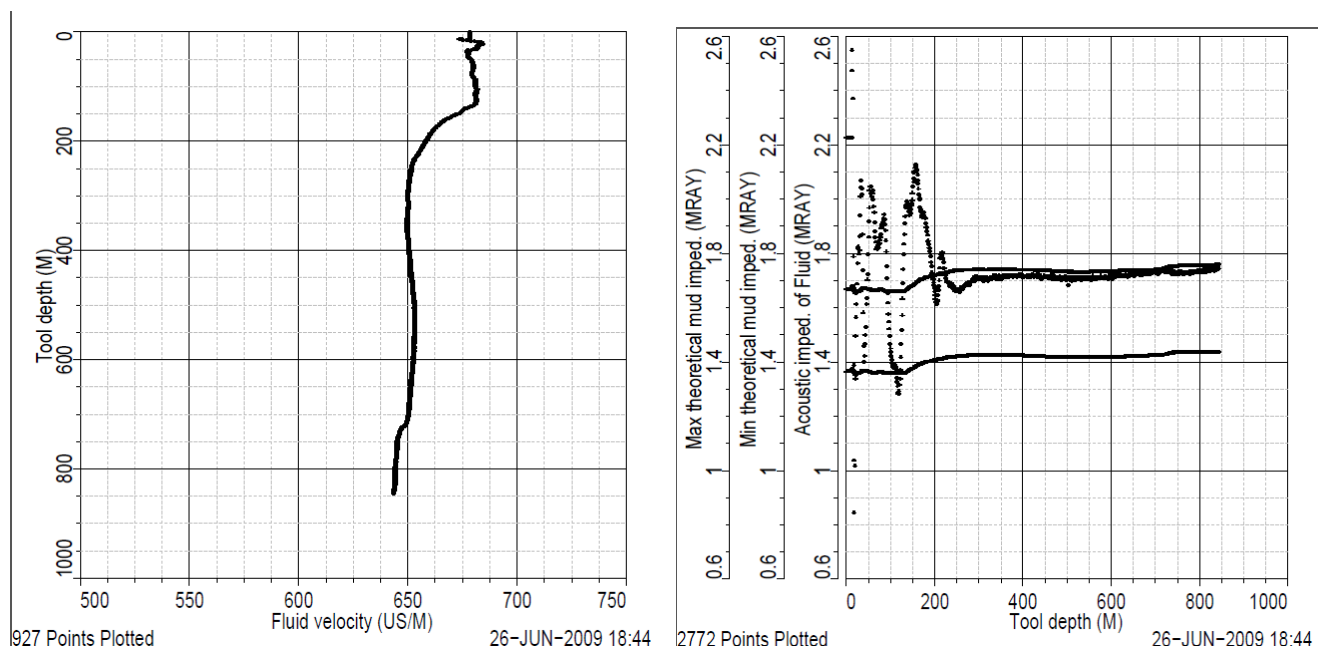


Figure 2 Fluid Properties Crossplots (not relevant since these values have not been used in the processing of the log)

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Zoning of Mud Parameters

Depth	Fluid Velocity (DFVL)	Acoustic Impedance (ZMUD)
900.00	196.61	1.60
700.00	198.14	1.60
400.00	198.64	1.60
200.00	200.57	1.60
100.00	207.66	1.60

Figure 3: Fluid properties using during the processing of the log

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2.2 USIT Log Quality

Tool Centralisation	☒	☐	☐	<i>Eccentralisation within tolerance < 0.27 in.</i>
First Echo Amplitude	☒	☐	☐	<i>Strong signal, nearly half way across the 0- 75dB scale</i>
Processing Flags	☒	☐	☐	
Radius Curves within Specification	☒	☐	☐	<i>Dominatly within specification. IRAV: 6.18 in. ERAV: 6.68 in. except for some ovalized parts.</i>
Thickness Curves within Specification	☒	☐	☐	<i>Within specification THAV:0.51 in.</i>
Manufacturing Patterns visible on thickness image	☒	☐	☐	
Repeatability	☒	☐	☐	
Agreement with CBL/VDL	☒	☐	☐	

2.3 CBL/VDL Log Quality

Transit Time as expected	☒	☐	☐	<i>As expected around ~400 US</i>
Free Pipe Amplitude as expected	☒	☐	☐	<i>Expected Free pipe amplitude: 47mV Reference Amplitude =47 , FCF=0.903, CBAF=1</i>
Tool eccentricisation (transit time variation)	☒	☐	☐	
Repeatability	☒	☐	☐	

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3 Interpreted Results

3.1 Pipe Integrity Summary

Internal pipe wear grooves	No	
Minimum remaining wall thickness	Like nominal	
Maximum % Wall Loss (computed based on nominal thickness)	none	
Indications of Casing Ovality	Yes	<i>See table 1, for the intervals with ovalization. And figure 5 for definition of slightly and strongly ovalized.</i>
Indications of internal corrosion	No	
Indication of external corrosion based on comparison between internal radius and thickness measurements	No	

Table 1: Ovalization observed in the USIT log

Depth mMD	Degree of ovalization	Min radius
214- 217	Slightly ovalized	
252-254	Slightly ovalized	
258- 261	Slightly more ovalized	
261-267	Slightly ovalized	
326- 329	Slightly ovalized	
341-347	Slightly ovalized	
386-389	Slightly ovalized	
417-421	Slightly ovalized	
425-433	Slightly ovalized	
465-468	Slightly ovalized	
518-523	Slightly more ovalized	
536-548	Slightly ovalized	
556-588	Slightly ovalized	
588-590	Slightly more ovalized	
632-634	Slightly ovalized	
709-720	Slightly more ovalized	
730-734	Slightly more ovalized	
810-825	Strongly ovalized	
838-843	Slightly ovalized	

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3.2 Cement Bond Quality

The evaluations are based on data presented by the files:

STATOIL_Volve_15_9-F-9_USIT FINAL PRINT_THEORETICAL K FACTOR_1.PDF

Interval Top (mMD)	Interval Bottom (mMD)	Cement or Formation or Combination of both	Cement or Formation Bond Quality	Potential for Hydraulic Isolation ^[1]	Length of Interval if Isolating Potential is 'High' (m)	Comments
Double casing. 30 inch csg. shoe @ 220.8 mMD						
150	178	Cement	Moderate to Poor	Low		Patchy cement with small to large interconnected liquid filled pockets.
178	188	Cement	Good to moderate	Medium		This is in the 30" x 13 3/8" annulus – some isolating potential
188	195	Cement	Moderate	Medium		Good bond on CBL/VDL. Moderate bond on USIT due to patchy cement distribution in the annulus.
195	201	Cement	Good to moderate	Medium		Some small liquid pockets which do not seem to be connected. (See figure 6) This is in the 30" x 13 3/8" annulus – some isolating potential
201	221	Cement	Poor	Low		Large interconnected liquid filled pockets with some patchy cement distribution in the annulus. (See figure 6)
Single casing. 13 3/8" Casing shoe @ 909.8 mMD						
221	332	Cement	Poor	Low		Large interconnected liquid filled pockets with some patchy cement distribution in the annulus.
332	341	Cement	Moderate to Poor	Low		Cement on the high side with a channel running in the annulus.
341	440	Cement	Moderate to poor	Low		Very patchy cement with interconnected pockets of liquid in the annulus.
440	454	Cement	Moderate	Medium		Good bond on CBL/VDL. Moderate bond on USIT with small interconnected liquid pockets in between.
454	468	Cement	Moderate to poor	Low		Small interconnected liquid filled pockets with very patchy cement distribution in the annulus.
468	480	Cement	Moderate to poor	Medium to low		Large interconnected liquid filled pockets with a patchy to more homogeneous

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Interval Top (mMD)	Interval Bottom (mMD)	Cement or Formation or Combination of both	Cement or Formation Bond Quality	Potential for Hydraulic Isolation ^[1]	Length of Interval if Isolating Potential is 'High' (m)	Comments
						<i>cement distribution in the annulus.</i>
480	485	Cement	Good to moderate	Medium		<i>Some isolating potential</i>
485	513	Cement	Moderate	Medium		<i>Good bond on the CBL/VDL. USIT showed a narrow high side liquid filled pocket</i>
513	550	Cement	Moderate	Medium		<i>Good bond on CBL/VDL, some interconnected small liquid filled pockets</i>
550	675	Cement	Moderate	Low		<i>Cement with a channel is running in the annulus.</i>
675	705	Cement	Moderate to poor	Low		<i>Channel and patchy cement in the annulus.(cement seems less dense than the interval above)</i>
705	737	Cement	Moderate to poor	Low		<i>Channel and less patchy cement in the annulus.(compared to the interval above)</i>
737	751	Cement	Moderate to poor	Low		<i>Channel and less dense cement in the annulus.(compared to the interval above)</i>
751	757	Cement	Good to moderate	Medium		<i>Some isolating potential</i>
757	839	Cement	Moderate	Low		<i>Moderate bond on CBL/VDL. Moderate azimuthal bond on the USIT; channel.</i>
839	845	Cement	Moderate	Low		<i>Good bond on CBL/VDL. Moderate azimuthal bond on the USI; channel.</i>

Notes:

- [1] **High** – Implies the logs show cement or formation material with isolating potential that would satisfy Statoil's barrier verification requirements
Medium – Implies some sealing potential but with some uncertainty such that it cannot be confirmed with logs alone
Low – Implies no real sealing potential
- [2] Please note that for Formation Bond the current requirements state that a **50m continuous interval** is required for the shale formation to be verified as an annular barrier element.
- [3] For formation bond classified with a 'high' sealing potential to be considered as a barrier it is assumed that this formation has previously been tested and verified and meets the WBEAC criteria as described in the ARIS Well Integrity Manual.

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4 Appendix

A.1 Figures

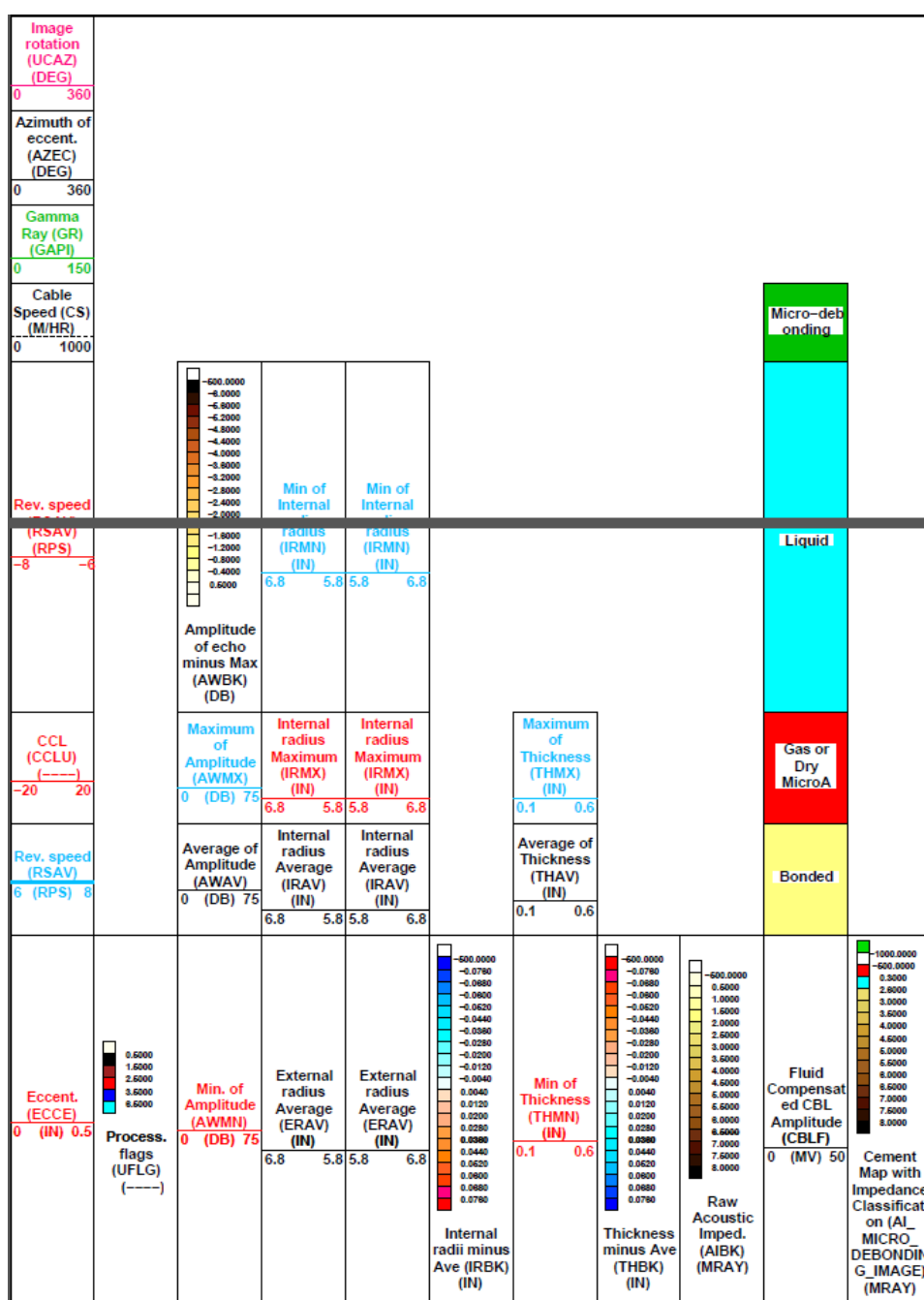


Figure 4: Heading for the log sections displayed below.



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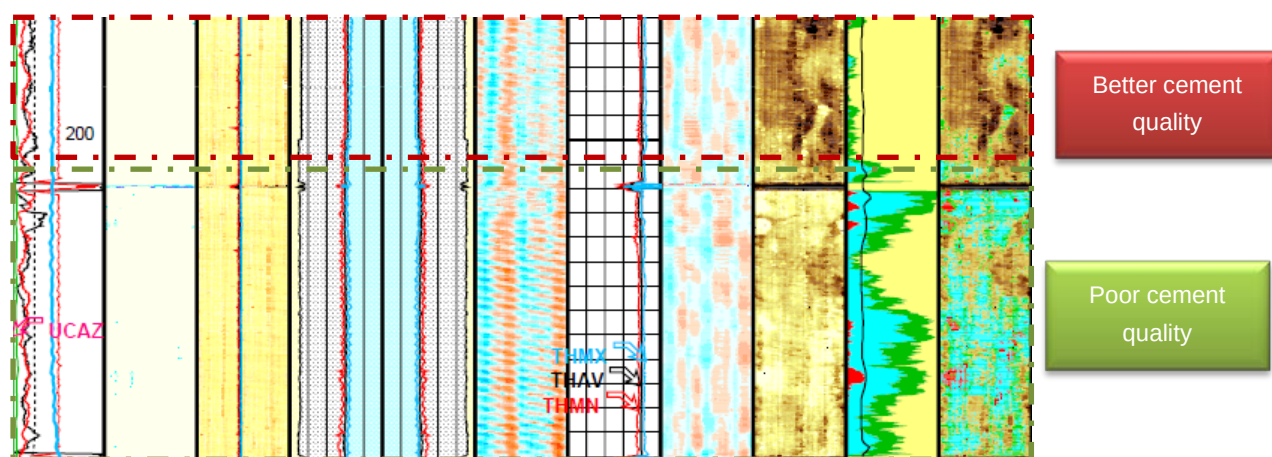


Figure 6: Example Log Section 195- 213 mMD, displaying a good and poor cement interval

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A.2 Schlumberger Sequence of Events (SOE)

Sequence of Events	
Well: 15/9-F-09 Rig: Mærsk Inspirer Field: Volve Client: StatoilHydro ASA	
Engineers: (D) John Walsh (23-Jun-2009 / 29-June-2009) (N) Girish Chhatwani (24-Jun-2009 / 29-June-2009) (N) Arjun Bhasker (N) (23-Jun-2009 / 29-June-2009)	
Operators: (D) Jarl Henning Aareskjold (23-Jun-2009 / 29-June-2009) (N) John Bjoreland (23-Jun-2009 / 29-June-2009)	
23-June-09 =====	
20:30	Looking for the unit, doghouse and containers
21:30	Find the rig electrician to power up the unit
23:00	Unit is powered up and started purging the unit
23:35	Purging up finished and started making connections inside the unit
24-June-09 =====	
00:30	Troubleshooting MCMPs and monitors
02:00	Inventory of the dog house
03:00	Getting Wellsite data and other information
08:00	Begin lifting tools/equipment into place
10:00	Fixing USB hubs
11:00	Replace TGRPs
13:00	Replace malfunctioning MCMP
14:10	Mount drum on Seawell winch
15:00	Collector connected
19:00	Getting work permits for megging the cable/ head and op-checking tools on the pipe deck
20:30	Connecting the umbilicals to the winch
21:00	Setting up TDC for calibration of the CMTD
22:00	Checking cable continuity and performing insulation checks
	Line 1 : 286 Ohms
	Line 2 : 283 Ohms
	Line 3 : 284 Ohms
	Line 4 : 280 Ohms
	Line 5 : 285 Ohms
	Line 6 : 284 Ohms
	Line 7 : 258 Ohms
22:30	Calibration of the CMTD using a TDC
23:30	Mounting IDW and CMTD on the Seawell winch
25-June-09 =====	
00:30	Checking IDW and making connections
01:00	Started connecting Main toolstring for operational checks
02:00	Started checking Main toolstring
09:00	Finish checking main toolstring
10:30	Connected backup toolstring
11:00	Client meeting
13:30	Finish checking backup toolstring
19:30	Client meeting
20:30	Troubleshooting problems with monitor, air conditioner
26-June-09 =====	
13:00	Working on Presentations, Interact Access, FTL Tickets, Parameters, Zebra outputs, DQA
14:20	Toolbox meeting with the rig crew
14:45	Started rigging up
16:00	Tool powered up and surface checks performed
16:15	Surface checks completed and started running in hole
16:21	Log down started - 012.DLIS and continued running in hole
17:59	Log Up started - 014.DLIS
	DO: +0.3m
21:30	Reached 25m but due to poor data in the riser section, went down to 170m to get the data again with different parameters from the Sub sea well head to the surface - 016.DLIS
	DO= -0.1m
22:30	Reached surface again and started running in hole again for the repeat pass
23:26	Started repeat pass from 845m - 022.DLIS3
	DO = +0.3m
27-June-09 =====	
03:16	Tool at surface and ready to rig down, waiting on permission to rig down
03:30	Started rigging down.
04:30	Completed rigging down and started packing tools and equipment
08:00	Packing equipment and doing doghouse inventory

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A.3 Tool drawing

